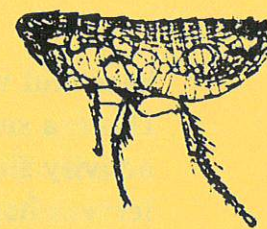


flea NEWS

40



Bernard C. Nelson

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EDITORIAL

Under the banner **A major museum goes populist**, the May 3 issue of the prestigious British magazine **NATURE** carried a two-page editorial concerning the state of affairs at the British Museum (Natural History). Its lead-in stated "Yet another British institution is in the throes of ill-calculated change: the British Natural History Museum will subordinate research to popular exhibition. It should rather have rejected the task imposed on it." The author goes on to outline the effects the decades of chronic penury have had on the Museum and its programs, including shockingly low salaries for its staff and impaired physical maintenance. Since 1987 the Museum has charged admission, and another article in the **New Scientist** (28 April 1990) by Mike Hamer points out that earlier this year these charges rose from £2 to £3 per person.

It seems the matter came to a head when Director Neil Chalmers released the Museum's 1990-95 "corporate plan" on April 23rd. In an analysis by Henry Gee published in **NATURE** on April 26th, the effect of implementation of the plan would be to reduce the staff by one out of every six positions by 1992-93 and reduce the emphasis on research while increasing the exhibit program to make the Museum more attractive to the general public. Six research areas have been designated and include biodiversity, environmental quality, living resources, mineral resources,

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human health and human origins. While these would appear to be deserving categories at first glance, I suspect that they better serve the administrative and political factions of the institution than the Science that the institution was created to steward.

In a subsequent article in **NATURE** on May 3rd under the heading **Taxonomy pays for bad image**, Gee reports that researchers at the Natural History Museum went on a one-day strike to protest against the "corporate plan", which proposes the loss of 51 of 300 research and curatorial positions during the next two years. Of the five departments, listed proposed reductions are as follows: Botany, 7; Entomology, 17; Mineralogy, 10; Palaeontology, 10; and Zoology, 7. Five of the positions in Entomology would be lost through curatorial reorganization. The remaining 12 would be in research areas on insect sounds, functional morphology in beetles, weevils, mayflies, lacewings, caddisflies, heteropteran & collembolan bugs,

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bees and wasps, and microlepidoptera. Fleas, now included in the Medical Diptera section, are not mentioned, probably because there has been so little activity since the retirement of F. G. A. M. Smit in June of 1980. It is our fervent hope that the benign neglect enjoyed by the flea collection for the past 10 years will not be replaced by malign neglect in the future. - R.E.L.

BERNARD C. NELSON

2.VI.1934 - 18.II.1990

We are deeply saddened to report the death of Bernard C. Nelson, Ph.D., who passed away at Mt. Diablo Hospital, Concord, California, after a long and courageous battle with cancer.

Bernie was born at Cass Lake, Minnesota. He received his Bachelor of Science degree from the University of Wisconsin in 1956 and a Master of Science from the University of Michigan in 1960. He earned a Doctor of Philosophy degree in parasitology from the University of California at Berkeley in 1967.

He was awarded a National Institutes of Health Fellowship in 1968 and studied the ecology of avian lice during the next two years at the McMaster Animal Health Laboratory in N.S.W., Australia. On his return from Australia in 1971, he joined the California Department of Health Services, Environmental Management Branch, as a parasitologist. In 1978 he became head of the Zoonoses Unit where he pursued a distinguished and productive career in the control and prevention of plague and other vector-borne diseases, bringing his Unit into national and international prominence.

Bernie was a dedicated and enthusiastic worker, whether in the field with rodents, fleas and plague, or in the laboratory on flea taxonomy. His conceptual thinking on the dynamics of plague and plague ecology, especially in California, was unparalleled. Working with Bernie was a joy; he had the knack of making a hard day's work pass quickly and a dull day exciting.

We will miss him!

-Allan M. Barnes

RECENTLY PUBLISHED

A Catalogue of Invalid or Questionable Genus-Group and Species-Group Names in the Siphonaptera (Insecta). Lewis, R. E. & J. H. Lewis, 1989. ISBN 3-87429-302-5. Koeltz Scientific Books, D-6240 Koenigstein, FRG. 263 pp., 2 figs.(unnumbered). DM 120.00, US \$75.00 (available in the U.S.A. through Lubrecht & Cramer, Ltd., R.D. #1, Box 244, Forestburgh, N.Y. 12777).

This compendium is an attempt on the part of the authors to supply supporting information about names used in the order Siphonaptera that are currently considered invalid for one reason or another. The mimeographed work by Smit & Wright, 1978 (*A list of code numbers of species and subspecies of Siphonaptera*) formed the basis for this treatment, since these authors included junior synonyms and a number of types of invalid names (i.e., *nomina nuda*, *nomina oblita*, junior homonyms, etc.) in their listing. Unfortunately, no supplementary information was included about the taxa or their history. This type of information can be most useful to researchers but frequently it is extremely difficult to trace, requiring as it does an extensive library (and no small amount of luck).

Most of the names involved are junior synonyms and the format we followed was varied only when other types of names were encountered. A typical entry lists the name (species, subspecies or genus) and its author, followed by the citation where the name was erected or applied. In the case of species and subspecies, the collection locality, host, date and collector are given, as well as type status and deposition when known. After variable annotation, most accounts end with the senior synonym or equivalent, and the citation where synonymy or other nomenclatural action was taken. In all, 670 names are dealt with and 639 citations are included in the References Cited. Although the cover indicates 1990 as the publication date, the actual date of publication is 31.XII.1989.

The work concludes with three appendices. Appendix I lists all of the valid avian and mammalian host names, with their authors and citations. Host synonymy is dealt with directly in the text of the taxon accounts. Appendix II lists the standardized abbreviations of the journals cited in the text with their unabridged equivalents. Appendix III is a short glossary in which nomenclatural terms are defined as they are used in the text.

As experienced systematists know, the validity of species and subspecies can be very much a matter of opinion among workers in a particular group

of organisms, and criteria employed for discrimination of taxa in one group of insects may be worthless in another group. Since experience is so important, where possible we have depended upon the work of others, avoiding the temptation to become judgmental ourselves with the exception of *Chiropteropsylla brockmani johnsoni* Hubbard, 1956, *Ctenophthalmus costai libanensis* Lewis, 1970, *C. bifidatus mancus* Beaucournu & Orhan, 1986 [more about this taxon elsewhere], *Ctenocephalides arabis* *multispinosus* Smit, 1960, and *Paraceras melis sinensis* Sakaguti, 1962, all of which have been relegated to their nominate taxa.

No work of this type can ever be entirely complete, nor can it be completely error-free. We have requested colleagues who note errors of mechanics or fact to inform us in order that these may be included in future numbers of Flea News. The following is a critique submitted by Dr. J.-C. Beaucournu. It is translated from a letter dated 26.IV.1990. Comments in brackets ([]) are those of R.E.L.

Sarcopsylla gallinacea erinacea is perhaps a synonym of *Echidnophaga murina* (types lost).

Ctenophthalmus congener grulichii: I continue to think that it is a good (although perhaps a subtle) subspecies.

Parapsyllus magellanicus heardi: elevated to full species status in Smit, 1987.

Coptopsylla johanae[sic!]: placed in synonymy with *C. africana* (Launay & Beaucournu, 1987). It is at least questionable.

Ctenophthalmus bifidatus mancus: the criterion of distinction is not the phallosome but segment IX and in particular st IX! (Beaucournu et Orhan, 1986, fig. 5). It should not therefore be synonymized with *C. bifidatus*. [The only male in my collection does indeed possess a well developed ventral projection from the base of the distal arm of st IX, and *C. b. mancus* could possibly be a good species. Additional material from the type locality is needed to resolve this problem.]

Hystrihopsylla narbeli: could be synonymized with *H. t. talpae* or a senior synonym of *H. t. alpina*; unfortunately the type is lost.

Neocoptopsylla: the generic level (category) is subjective, but it is manifestly

a taxon well separated from the others (*N. wassiliewi*): then genus, subgenus.....? I do not think that the retained bristle character is minor.

Citellophilus occidentalis: the place of capture is Gave d'Ossoue and not Gave d'Ossone. It is a *lapsus calami* of Jordan. See, in particular, Beaucournu, 1969 (1968) - *C. [tenophthalmus] nivalis ianlinni*.

Rhadinopsylla pitymydis is considered (wrongly or correctly) as a synonym of *R. isacantha* (Launay & Beaucournu, 1982(1983)). Again, a species at most "questionable".

Myoxopsylla laverani traubi = *M. laverani* (I have your letter on this subject) (Beaucournu & Alcover, 1984). [Although it is still treated as a valid taxon in an upcoming classification of the Ceratophyllidae, I am inclined to agree that it is probably equal to the nominate taxon.]

-R.E.L. & J.-C.B.

Les Puces (Siphonaptera) de France et du Bassin Méditerranéen Occidental. Beaucournu, J.-C. & H. Launay, 1990. Faune de France 76. 548 pp., 1000 figs. + 1 photomicrograph, 102 maps. ISBN 2-903052-10-7.

This is the first comprehensive treatment of the flea fauna of western Europe and, in addition to France, includes Italy, Switzerland, Spain, Portugal, the western Mediterranean islands west of Sicily and Malta, the Canary Islands, and Morocco, Tunisia and Algeria in its coverage. In all, 213 taxa are included, belonging to 138 recognized species.

The introductory portion of this work includes a brief history of the study of fleas, the external morphology of both the larvae and adults, as well as a brief review of the internal anatomy of the adult, a consideration of flea biology and host specificity, a section on phylogenetic affinities, classification and a brief discussion of some taxonomic problems resident in the fauna of southern France and the Iberian Peninsula, and terminates with sections on the role of fleas as vectors of pathogens, pathology of the fleas themselves and techniques for the collection, preservation and study of these fascinating insects.

The bulk of the book deals with the systematic accounts of the taxa and follows the following general format: synonymy, if any; type deposition, if known; iconography; morphology; distribution; host specificity; ecology; and references. All taxa are illustrated by line drawings and distribution

maps are included for most taxa.

The systematic accounts are followed by four appendices. Appendix 1 is a host index where all of the flea taxa reported are listed either as primary or secondary parasites of the host in question. Birds and mammals are treated separately. Appendix 2 is an alphabetical index of flea species and subspecies, including synonyms. Appendix 3 is an alphabetical subject index, and Appendix 4 is a glossary of anatomical terms employed in the keys and the text. The book terminates with an exhaustive bibliography and a Table of Contents.

The volume is bound in soft covers and bears a particularly spectacular color photomicrograph of an *en face* view of a cat flea taken on a Scanning Electron Microscope.

This is the culmination of studies performed by the authors and their colleagues over the past three decades, as well as an expansion of a doctoral dissertation by the senior author delivered in 1976. Although I am not in agreement with some of the more subjective taxonomic interpretations espoused by the authors, the work is well done and vital to students of the European flea fauna. The authors are to be congratulated for their devotion to the subject.

Following is my commentary on the nomenclatural actions taken by the authors in this work.

Coenopsylla. In: Roubaud, E. 1928. *Les Faunes régionales des Rongeurs et des Puces dans leurs rapports avec la Peste; Résultats de l'enquête du Comité permanent de l'Office international d'Hygiène publique 1924-1927.* Masson et Cie éd., Paris. 306 pp. We have not seen this publication.

= *Caenopsylla* Rothschild, 1909. (Beaucournu & Launay, 1990: 364. Said to be a *lapsus calami*.)

corsus Peus, 1963 - **Ctenophthalmus agyrtes**

Beitr. Ent. 13: 435, figs. 4a-c.

Corsica, Vezzani, 42.11N 09.10E, from *Mus musculus*, 21.IV.1953, HOLOTYPE male. One male paratype from Corte, 42.19N 09.07E, Restonica-Tal, from *Crocidura russula*, 17.IV.1952. Both retained in the Peus Collection.

= *Ctenophthalmus (Ctenophthalmus) agyrtes sardiniensis* Ioff, 1928. (Beaucournu, Rault & Beaucournu-Saguez, 1965, Bull. Soc. Zool. Fr. 89:750.)

ferox Fabricius - **Pulex terrestris**

In: Alpinus [=Faige-Blanc], 1874, *La Chasse alpestre en Dauphiné*. Baratier fr. et Dardelet éd., Grenoble, 395 pp. Cited in Beaucournu & Launay, 1990: 28, 52, as a *nomen ex phantasia*.

heselhausi (Oudemans, 1914) - **Ctenophthalmus bisoctodentatus**

Ent. Ber., Amst. 4: 139

Netherlands, Valkenburg, 50.52N 05.50E or 52.11N 04.26E, from nest of *Talpa europaea*.

= *Ctenophthalmus (Ctenophthalmus) bisoctodentatus bisoctodentatus* Kolenati, 1863. (Beaucournu & Launay, 1990: 256.)

musculi Bouché, 1835 - **Pulex**

Nova Acta physico.-med. 17: 508

This is a junior homonym of *Pulex musculi* Dugès, 1832, and a junior synonym of *Pulex* [= *Leptopsylla*] *segnis* (Schönherr, 1811).

opistolatus Hastriter, 1973 - **Ctenophthalmus andorrensis**

Erected in a thesis for the degree of Master of Science, but never formally published. Mentioned in synonymy by Beaucournu & Launay, 1990: 228.

= *Ctenophthalmus (Ctenophthalmus) andorrensis veletensis* Beaucournu, Gilot & Vericad, 1973.

pitymydis Zavattari, 1914 - **Neopsylla**

Annali Mus. civ. Stor. nat. Giacomo Doria 46: 144

Italy, Liguria, Cascinelle, Borzoli, from *Pitymys savii* and *P. multiplex* [= *subterraneus*], 5.IV.1910 and III.1909.

= *Rhadinopsylla isacantha* (Rothschild, 1907). (Launay & Beaucournu, 1982: 472.)

savignyi J. & R. - **Ctenophthalmus**. In: Roubaud, E. 1928. *Les Faunes régionales des Rongeurs et des Puces dans leurs rapports avec la Peste; Résultats de l'enquête du Comité permanent de l'Office international d'Hygiène publique 1924-1927*. Masson et Cie éd., Paris. 306 pp. A *lapsus calami* for *Ctenophthalmus (Euctenophthalmus) savii savii* J. & R., 1921, according to Beaucournu & Launay, 1990: 285.

sintrensis Ribeiro & Capela, 1988 - **Typhloceras favosus**

III Cong. Ibér. Entom., Granada, 256, figs. 1-4

Portugal, Estremadura province, environs of Sintra, 38.48N 09.22W, from *Crocidura* sp. Listed in the synonymy of *Typhloceras favosus rolandi* Jordan, 1938, by Beaucournu & Launay, 1990: 139, preceded by a ? However, it is treated as valid on page 144. Although two papers by Ribeiro & Capela are listed in the bibliography as being published in 1985, the page 139 citing gives 1987 for the date, while that of page 144 gives the date as 1988. The latter is correct.

tangerensis Hastriter & Tipton, 1975 - **Ctenophthalmus russulae**
J. Egypt. Publ. Hlth Assn 50(2): 138

Morocco, Taza province, 23 km S Taza, ca. 33.55N 04.01W, from *Apodemus sylvaticus*, 16.VI.1970, M. G. Hearst. HOLOTYPE male, USNM No. 7211. Allotype female from Tanger province, 3 km S Cape Spartel, ca. 35.50N 05.55W, 15.V.1970, L.W.Robbins.

= *Ctenophthalmus (Medioctenophthalmus) russulae galloibericus* Beaucournu & Lumaret, 1972. (Beaucournu & Alcover, 1984: 381.)

tenerifensis Gil Collado, Rodriguez Rodriguez & Zapatero Ramos, 1982 -
Echidnophaga

Revta iber. Parasit. (Vol. Extra): 102, figs. 1-11

Spain, Canary Islands, Tenerife Island, 28.15N 16.35W, from *Rattus norvegicus*, P. G. Santiago, 4 male, 13 female syntypes, deposited in the Parasitology Department, University Complutense, Madrid.

= *Echidnophaga gallinacea* (Westwood, 1875). (Beaucournu *et al.*, 1989: 49)

thoracicus Rothschild, 1911 - **Ceratophyllus**
Annls Sci. nat., Zool. (9)12: 210, figs. 3-4

Norway, Kirkenes, 69.41N 30.00E, Finnmark, no host recorded, 21.VI.1881, G. Pouchet, MHNP. Listed as *Amphipsylla thoracicus* in Jordan & Rothschild, 1920: 13, 98, as a junior synonym of *A. sibirica* by Wagner, 1930, and Smit & Wright, 1978a, and as a junior synonym of *A. sibirica sepifera* Jordan & Rothschild, 1920, by Beaucournu & Launay, 1990: 348.

traubi Lewis, 1966 - **Myoxopsylla laverani**
J. Parasit. 52(6): 1167

Egypt, Matruh governorate, Ageeba, 28 km W Marsa Matruh, ca. 31.22N 27.55E, from *Eliomys quercinus*, 9.XII.1964, H. Hoogstraal, HOLOTYPE male, USNM No. 69161. First assigned as a junior synonym of

the nominate taxon by Beaucournu & Alcover, 1984: 388. Repeated in Beaucournu & Launay, 1990: 391.

= *Myoxopsylla* (*Myoxopsylla*) *laverani* (Rothschild, 1911)

undecimdentata (Kol.) - **Tarsopsylla**. In: Roman & Pichot, 1974: 51. A *lapsus calami* for *Tarsopsylla octodecimdentata octodecimdentata* (Kolenati, 1863) according to Beaucournu & Launay, 1990: 387.

The following emended spelling is introduced on page 232.

boisseaui Beaucournu, 1968 - **Ctenophthalmus baeticus**

Bull. Soc. Scient. Bretagne 42: 241.

Emended to *Ctenophthalmus* (*Ctenophthalmus*) *baeticus boisseauorum*.

Finally, the following names are treated as junior synonyms, an action with which I currently disagree. *Nosopsyllus* (*Gerbillophilus*) *henleyi mauritanicus* (J. & R., 1912) and *Nosopsyllus* (*Gerbillophilus*) *henleyi israelicus* Smit, 1960, are both synonymized with *Nosopsyllus* (*Gerbillophilus*) *henleyi* (Rothschild, 1904).

Coptopsylla joannae Lewis, 1964. This name was originally synonymized with *Coptopsylla africana* Wagner, 1932, by Launay & Beaucournu, 1987, Ann. Parasit. hum. comp. 62: 159, 166, 170, and the synonymy is repeated here. A long series in my collection from Saudi Arabia, Jordan, and northern Syria convinces me that the species is distinct. -R.E.L.

Manual of Nearctic Diptera. Volume 3. J. F. McAlpine (Ed.), 1989. Monograph 32. Agriculture Canada, Ottawa, Ontario. i-vi, 1333-1581, figs. 114.1-116.9, tables 116.1-116.10. ISBN 0-660-12961-2 (v. 3). US\$ 91.15.

This is the third of the incomparable series of volumes published by our dipterist colleagues in Agriculture Canada, dealing with the flies of the Nearctic Region. That it should be mentioned here lies in the generally accepted evolutionary theory that fleas and the order Diptera share some type of common ancestry. The subject is explored on pages 1336-1339 of Chapter 114 - Phylogeny and Classification of the Nematocera by D. M. Wood and A. Borkent. The authors discuss the debate over whether or not the suborder Nematocera is a monophyletic group of lower dipteran families and opine that it is probably paraphyletic. They also briefly touch on the debate about whether the Siphonaptera is a sister group of the

Diptera (*cf.* Boudreaux, 1979), the Mecoptera (*cf.* Mickoleit, 1981 & Hennig, 1981), only the mecopteran family Nannochoristidae (*cf.* Imms, 1944 & Tillyard, 1935) or the Mecoptera plus the Siphonaptera (*cf.* Kristensen, 1975). They go on to propose an equally valid alternative to those listed above that places the Siphonaptera, Diptera and Nannochoristidae as sister groups, excluding the remaining mecopteran families based on an analysis of the following six character states involving larvae, pupae and adults.

Larvae:

1. Presence of an egg burster on dorsum of head capsule in first instar (plesiomorphic) or absent (apomorphic)
2. Eye with convex, lens-like cuticle (plesiomorphic) or of group of pigmented cells under transparent, unmodified cuticle (apomorphic)
3. Thoracic legs present (plesiomorphic) or vestigial or absent (apomorphic)

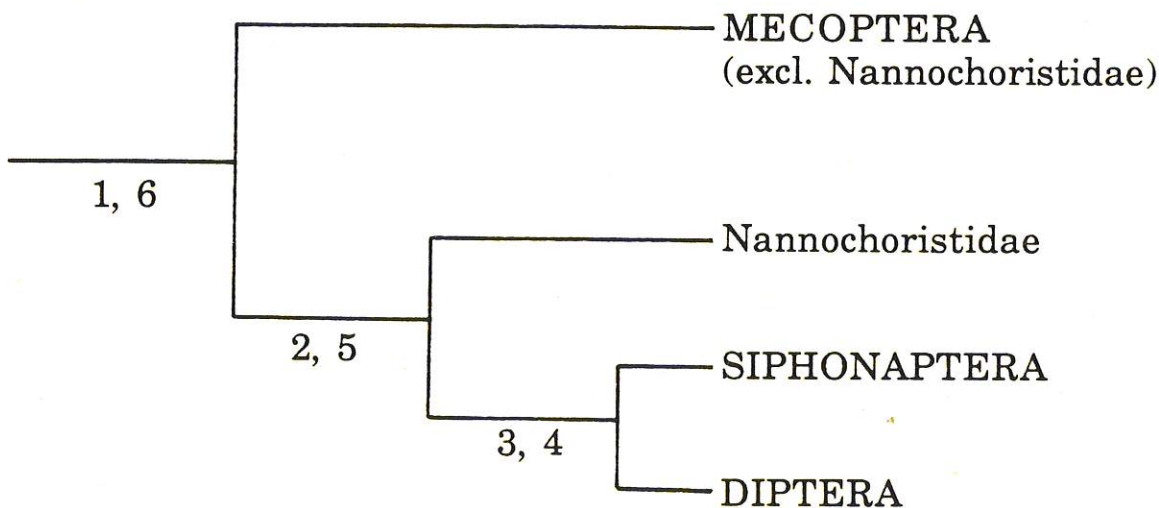
Pupae:

4. Decticous (with movable mandibles) (plesiomorphic) or adecticous (mandibles not movable) (apomorphic)

Adults:

5. Lacinia and galea both present and blunt apically (plesiomorphic) or galea lost and lacinia long and blade-like with recurved teeth (apomorphic)
6. Sperm pump present (plesiomorphic) or absent (apomorphic)

Based on these, and other considerations, the following cladogram was proposed, where the numbers refer to character states.



The authors state "...that the Nannochoristidae is a sister group of the Diptera plus Siphonaptera rests on correct interpretation of homology..." which may be the single, most cogent criticism of cladistic methodology. Even so, earlier workers have noticed these evident affinities and Hinton (1981) has even proposed the ordinal name Nannomecoptera for the Nannochoristidae.

This volume terminates a masterful series of studies on Nearctic Diptera and is well worthy of the award bestowed upon it and its contributors at the annual meeting of the Entomological Society of America at San Antonio, TX in December of 1989. -R.E.L.

CONGRATULATIONS!

Congratulations to Dr. Mathias Kiefer who successfully defended his dissertation on June 12, 1989. His work is entitled "Fleas of Mongolia, Zoogeography and Medical Importance". Dr. Kiefer is currently (December 1989) in Munich. (See page 363)

Terry Galloway of the University of Manitoba reports that Leslie R. Lindsay completed his thesis for the Master of Science in Entomology in December, 1989. Its title is "Patterns of seasonal activity and host-parasite relationships for fleas associated with Richardson's ground squirrels, *Spermophilus richardsonii*, in southern Manitoba, Canada".

NOTES

On page 336 of Flea News 39 we cited three names thought to have been published in 1987 and assigned species numbers to them accordingly. We have since been informed by Professor Beaucournu that the descriptions were not actually published until 1988. These three names should be transferred to the 1988 list and ultimately will be assigned numbers in that series. The names are:

Nosopsyllus (Gerbillophilus) maurus angustus
Nosopsyllus (Gerbillophilus) garamanticus
Delostichus incisus

The 12 January 1990 issue of SCIENCE (247:156) contains a brief discussion of work being done at Purdue University (West Lafayette, Indiana) on the effectiveness of ultrasonic flea collars (see Dryden *et al.*, 1989, 195: 1717). The conclusion was that "there is no evidence that fleas can either hear or produce ultrasonic signals".

The following have sent reprints and other literature since Flea News 39. Many thanks! D. Beard, K. Bartkowska, D. Cyprich, E. Easton, T. Galloway, R. George, Gong Z.-d., R. Haitlinger, Liu Q., J. Müller, R. Pilgrim, R. Rausch, V. Sapegina, M. Stanko, Wu H.-y.

CONFERENCES

The following presentations dealing with fleas were given at the annual meeting of the Entomological Society of America at San Antonio, Texas, December 10-14, 1989.

Daily egg and feces production by the cat flea (*Ctenocephalides felis*) on the host. -Kern, W. H., Jr., P. G. Koehler, N. C. Hinkle & R. S. Patterson

Estimating cat flea (*Ctenocephalides felis*) numbers on the host. -Hinkle, N. C., P. G. Koehler, W. H. Kern, Jr. & R. S. Patterson

A comparison of the ovicidal activity of (S)-methoprene against *Ctenocephalides felis* *in vivo* and *in vitro*. -Donahue, W. A., R. Young & R. W. Meola

Fire ant predation on the cat flea: a laboratory study. -Palma, K. G. & R. W. Meola

Blood volume uptake and reproductive efficiency of the cat flea, *Ctenocephalides felis felis*. -Dryden, M. W. & S. M. Gaafar

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MEDICAL ENTOMOLOGY UPDATE

Recent issues of the Technical Information Bulletin of the Defense Pest Management Information Analysis Center have reported two plague outbreaks in Africa. In Madagascar, more than 250 confirmed and suspected cases were detected in January & February, some being reported from the capitol, Antananarivo. In Botswana, 105 cases had been reported from villages up to January and 9 of them were fatal. Rodent and flea control measures seemed to be slowing the incidence.

TIB also reported 2 flea-borne murine typhus cases from Israel in 1989 but noted that overall incidence of vector borne diseases in this country was low.
-J.H.L.



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ADDITIONS

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PROVISIONAL LIST OF TAXA DESCRIBED IN 1988

Amphipsylla apiciflata Liu, Xu & Li
 Paradoxopsyllus aculeolatus Ge & Ma
 Amphipsylla yadongensis Wang & Wang
 Nosopsyllus (Gerbillophilus) maurus angustus Beaucournu & Launay
 Nosopsyllus (Gerbillophilus) garamanticus Beaucournu & Launay
 Delostichus incisus Beaucournu & Torres-Mura
 Plocopsylla lewisi Beaucournu & Gallardo
 Ctenoparia propinqua Beaucournu & Gallardo
 Nycteridopsylla liui Wu, Chen & Liu
 Callopsylla arcuata Ge, Wang & Ma
 Metastivalius novaehiberniae Beaucournu & Mahnert
 Mesopsylla sagitta Yu, Ye & Liu
 Ischnopsyllus (Hexactenopsylla) infratentus Wu, Wang & Liu
 Callopsylla (Callopsylla) longispina Zhang & Yu
 Pulicella aenigma Lewis & Cheetham
 Plocopsylla kasogonaga Schramm & Lewis
 Macrostylophora angustihamulus Li, Zhang & Zeng
 Ctenophthalmus longiprojiciens Chen, Li & Wei
 Coptopsylla lamellifer formozovi Darskaya
 Paradoxopsyllus angustisinus Cai & Wu
 Cratynius (Angustus) lui Gong & Lei
 Typhloceras favosus sintrensis Ribeiro & Capela

PROVISIONAL LIST OF TAXA DESCRIBED IN 1989

Palaeopsylla laxidigita Xie & Gong
Palaeopsylla polypina Xie & Gong
Palaeopsylla medimina Xie & Gong
Ctenophthalmus (*Palaeoctenophthalmus*) *harputus* Aktas
Ctenophthalmus solutus nasutus Beaucournu, Valle & Launay
Citellophilus simplex rosickyi Cyprich
Xenopsylla guancha Beaucournu, Alcover & Launay
Stenischia chini Xie & Lin
Stenischia liae Xie & Lin
Stenischia liui Xie & Lin
Stenischia wui Xie & Lin
Jellisonia amadoi Ponce-Ulloa
Jellisonia mexicana Ponce-Ulloa
Plocopsylla crypta Beaucournu & Gallardo
Rowleyella gongshanensis Gong & Duan

PROVISIONAL LIST OF TAXA DESCRIBED IN 1990

Hectopsylla pascuali Beaucournu & Alcover
Plocopsylla consobrina Beaucournu & Alcover
Agastopsylla boxi gibbosa Beaucournu & Alcover
Chiliopsylla allophyla tonnii Beaucournu & Alcover

NOTICE

Flea News is a semiannual newsletter devoted to matters involving insects belonging to the order Siphonaptera. It is compiled and distributed free of charge by Robert E. and Joanne H. Lewis with support from the Department of Entomology at Iowa State University in Ames, Iowa. It is mainly bibliographic in nature, but recipients are urged to check any citations cited here before including them in a publication. Many of our sources are abstracting journals and current literature sources such as **Current Contents®** and citations have not necessarily been checked for accuracy. Further, recipients are urged to contribute items of interest to the profession for inclusion herein.

-R.E.L.

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LITERATURE ON SIPHONAPTERA

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- Zhang Rong-guang & Yu Xin.** A new species of *Callopsylla* Wagner, 1934 from Gansu Province, China. (Siphonaptera: Ceratophyllidae) Endemic Diseases Bull. 3(3): 57-61, figs. 1-5 [*Callopsylla (Callopsylla) longispina* Zhang & Yu]
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- Gong Zhen-da & Duan Xing-de.** A new species of the genus *Rowleyella* from Yunnan, China. Endemic Dis. Bull. 4(4): 30-32, figs. 1-6. [*Rowleyella gongshanensis* Gong & Duan]
- Guiguen, C., A. M. Teixeira & J. Y. Monnat.** *Xenopsylla gratiosa* (Siphonaptera: Pulicidae) and *Ornithodoros maratimus* (Acarina, Ixodoidea, Argasidae), haematophagous ectoparasites of seabirds, new to Portugal. Revta iber. Parasit. 49(2): 177-182, 2 maps.
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- Kaczmarek, S.** Fleas from the nests of *Passer domesticus* L. and *P. montanus* L. VI Symposium on Medical and Veterinary Acaroentomology, p. 19.
- Kadulski, S.** Ectoparasites of the hare *Lepus europaeus* Pall. VI Symposium on Medical and Veterinary Acaroentomology, p. 20.
- Kessel, E. L.** Autobiographical anecdotes (I was a preacher's kid). Myia 4. xxix-xxxii, 1-184, Bound \$10.00. P. Arnaud (Ed.) [Edward Kessel wrote the 1939 classic study on flea embryology published in Smithsonian Misc. Coll. 98(3): 1-78, figs. 1-90. MYIA is an irregular series published by the California Academy of Sciences, Golden Gate Park, San Francisco, CA 94118.]

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- Stanko, M. & M. Ambros.** Beitrag zur Kenntnis der Ektoparasiten (Acarina, Siphonaptera) der Kleinsäugetiere (Insectivora, Rodentia) des Staatnaturschutzgebietes Ostrov Kopáč (Slowakei). Zbor. Slov. nár. Múz. Prír. Vedy 35: 91-100.
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- Beaucournu, J. C. & J. A. Alcover.** Pucés récoltée dans la province de Neuquén (Argentine): Description de 4 nouveaux taxa. (Insecta: Siphonaptera) Ann. Parasit. hum. comp. 64(6): 489-505, figs. 1-31. [*Hectopsylla pascuali*, *Plocopsylla consobrina*, *Agastopsylla boxi gibbosa* & *Chiliopsylla allophyla tonnii*Beaucournu & Alcover]
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"I don't understand all those signs for flea markets. Who could want fleas that badly?"

(Roger Simon of the **Baltimore Sun** in a column entitled **Serendipitous notions about life.**)

"Snodgrass states that he cannot understand how the genitalia function but, since fleas continue to reproduce effectively, clearly the male flea finds no such difficulty"

(From a lecture by the British Entomologist O. W. Richards.)

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